I-TIC1 Smart Controller

Compact Bluetooth Irrigation Control Unit 1 valve

APP Operating Manual

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1 I-TIC configurations

This control unit is basically available in two versions,

- compact design incorporating a bistable solenoid, the control unit can be screwed directly onto the valve
- separate execution with control of an external latching solenoid 9-12 Vdc (Tecnidro type ELBI or ELBA, or similar) connection via red-black cable

The control units in addition to the basic function of opening and closing the solenoid (integrated or separate) can optionally be equipped with:

- rain sensor
- meter reading with pulse output (typically reed contact)

2 Loading APP

The I-TIC smart valve APP is available in both iOS and Android environments and can be freely downloaded from the respective stores (AppleStore or PlayStore), referring to the APP

(Note: APP version ver. 1.03 or later compatible with I-TIC devices being distributed from April 2025, firmware ver. 3.1.5 and later)

Tecnidro I-TIC Smart Controller

When you start the APP on your smartphone/tablet, an essentially empty first screen appears where you can connect to a device.

On this page you can select the interface language ITALIAN or ENGLISH

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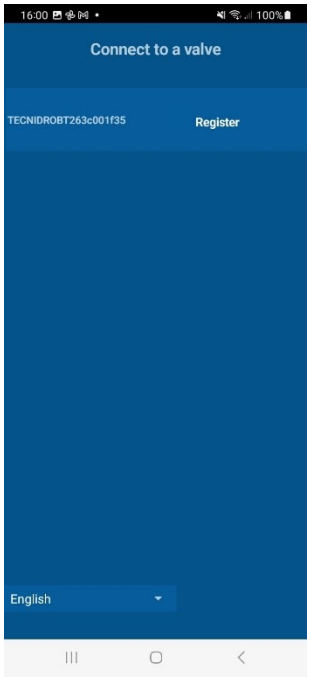
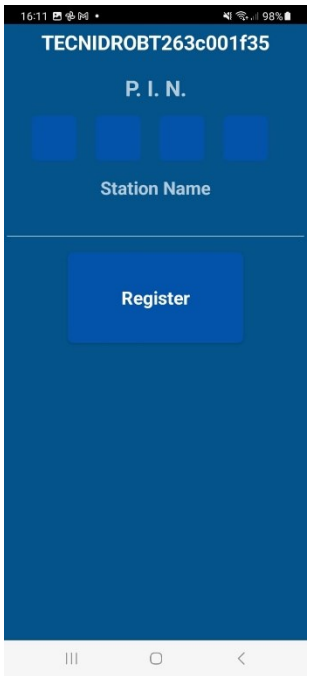
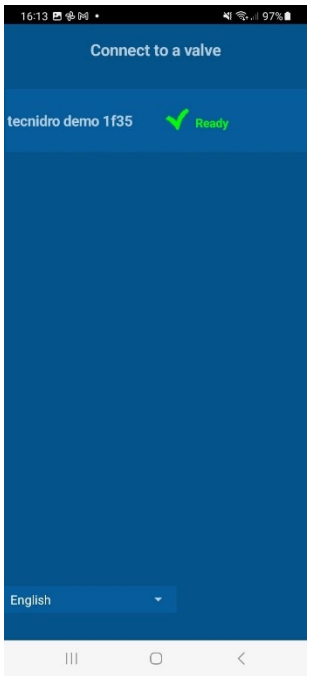


At this stage, Bluetooth scanning is active to 'find' any I-TIC devices to be paired.

2.1 Association with an I-TIC device

When the APP detects a device, it appears on the 'Connect to a Valve' page.





It is then possible to register the new unit by entering the PIN (required during registration defaut pin **1234**) and the station (device) name (free field, 3 to 20 characters). After completing registration by pressing the Register button, the unit is highlighted as **Ready**

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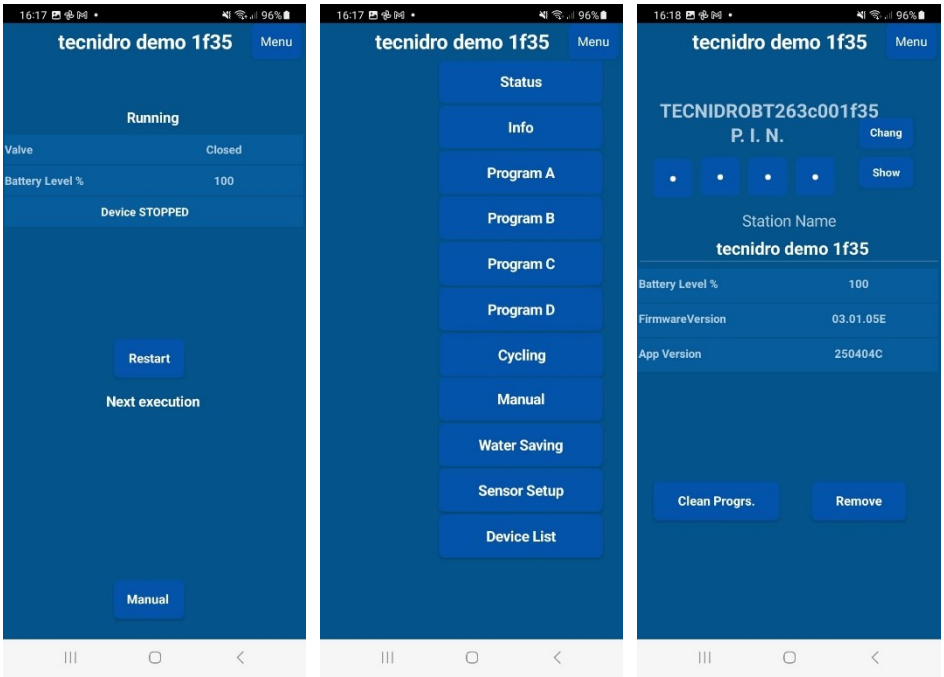
2.2 Change PIN

After registering a device this has the default PIN 1234, you can change the PIN by operating as follows:

- Select the device
- Select Menu
- Select Info



- This page shows the PIN and other information about the device
- The Show / Hide button allows you to show/hide the PIN
- The Chang button opens the PIN change page where the current PIN and the new PIN must be entered
- The Change button at the bottom of the page transfers the new PIN to the device and register it in the APP (an indication appears sending data to the valve)

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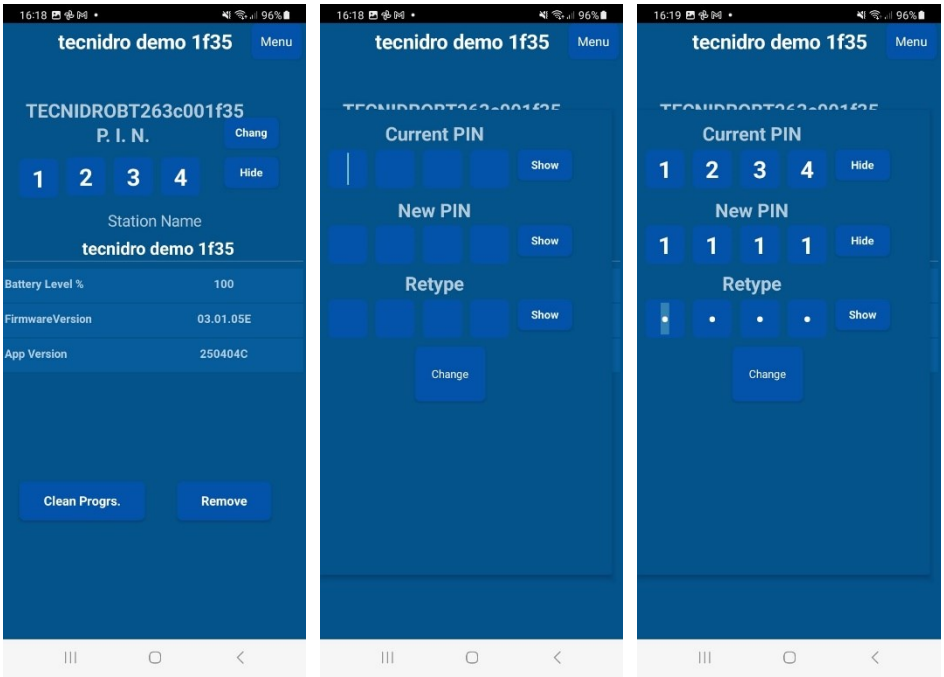

NOTE: It is possible to register the I-TIC on more than one smartphone device, to do this you must set the same PIN for the specific device.

2.3 PIN reset

Replacing batteries on the device results in resetting the PIN to the default PIN (1234), so the next time the APP is connected, it is necessary to reset the PIN itself on the APP.

This procedure is illustrated below by selecting a device, following connecting to the valve and data retrieving phase, a window appears indicating:

- Unable to connect. Maybe the pin is wrong. Do you want to change it?
- Select YES, the pin change window appears where the pin registered on the APP for that device is displayed (The Show / Hide button allows you to show/hide the PIN)
- Select Reset Pin, the factory pin reset window then appears
- Select YES, the APP confirms the reset and selecting OK returns you to the device list
- It is now possible to connect to the device (pin is 1234)

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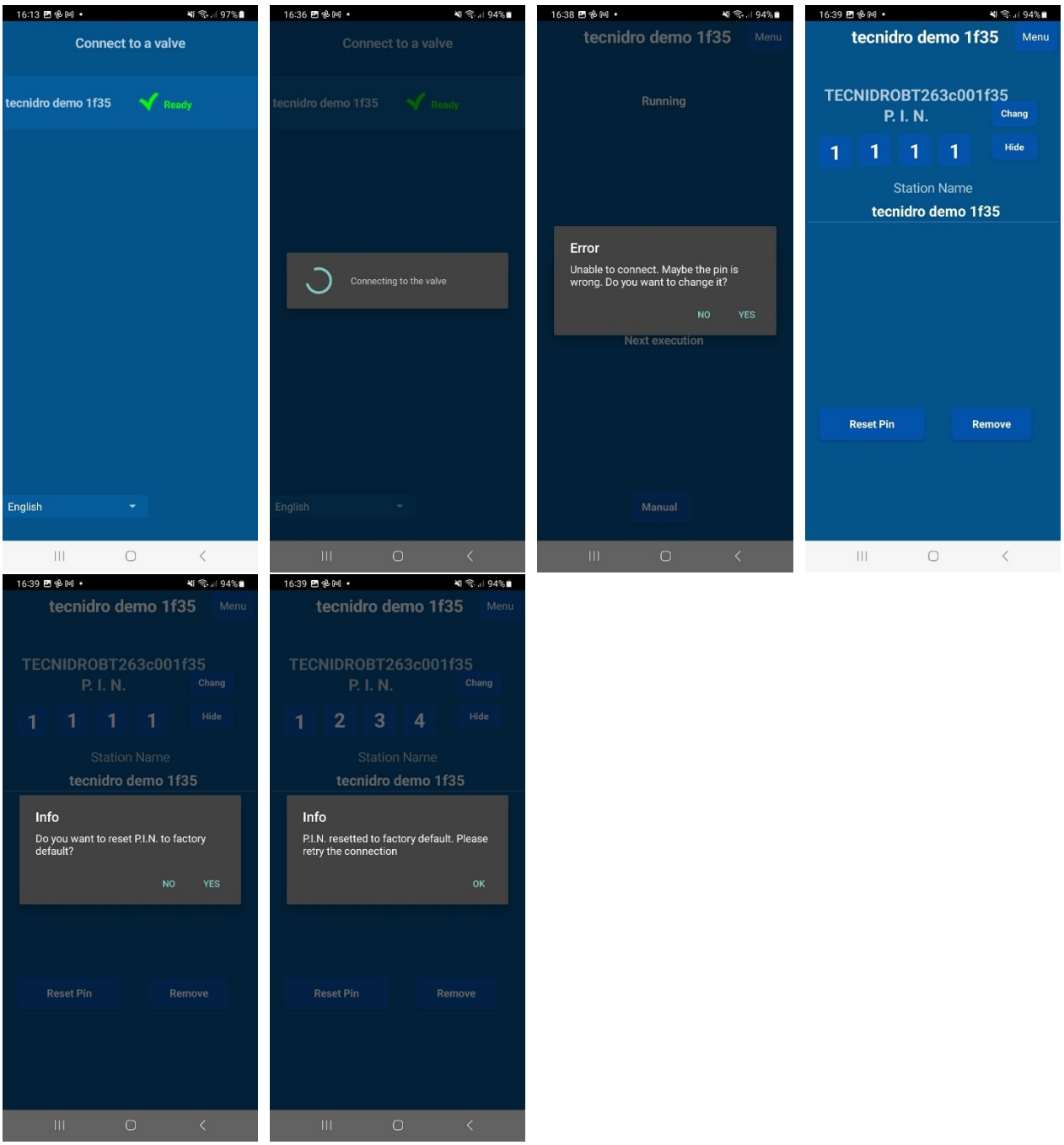

2.4 Deleting a device

If there is an unused device on the APP that you want to remove, you can delete it by operating as follows:

for example, in the device list there is the device “test 2fd6” that you want to delete

- Select the device you want to remove
- Select Menu
- Select Info
- Then select on Remove device and confirm with YES

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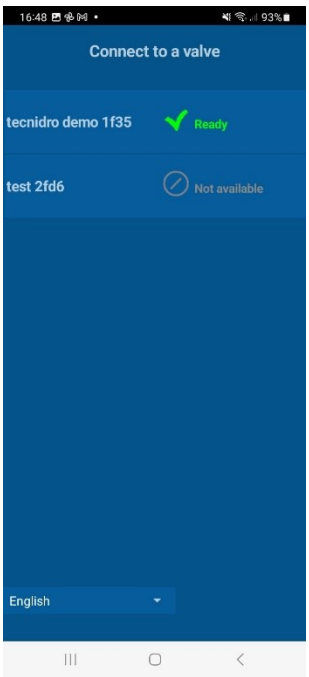

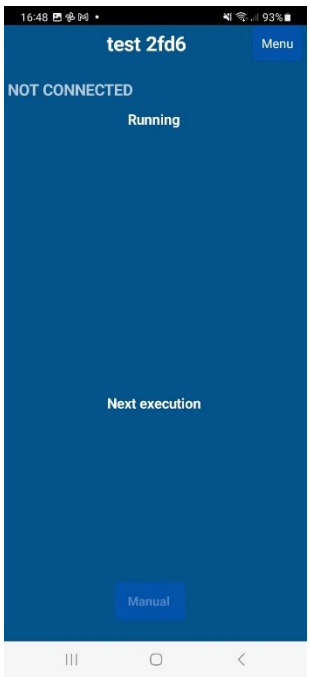

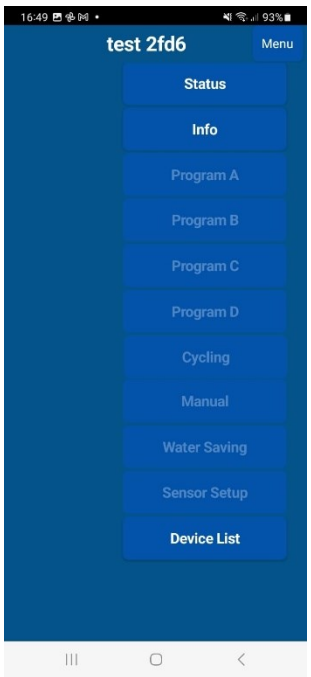

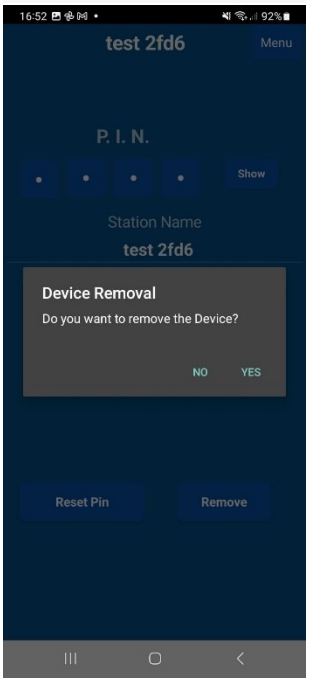

NOTE: You can delete both connected and unconnected devices at that time to the APP.

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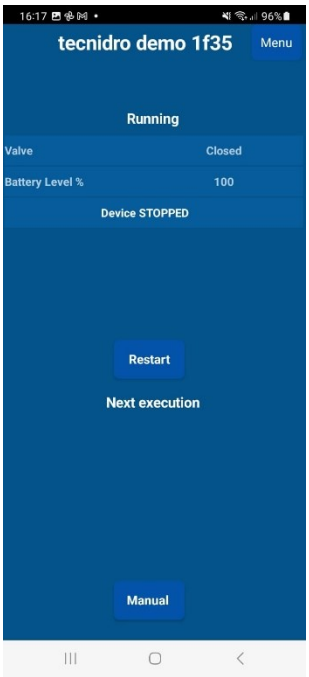
3 Device page

When accessing a device for the first time, the following page is displayed

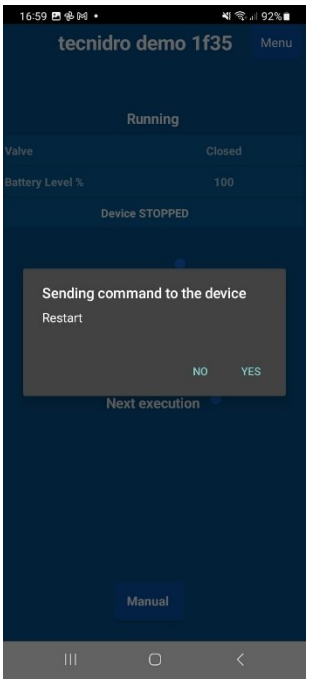


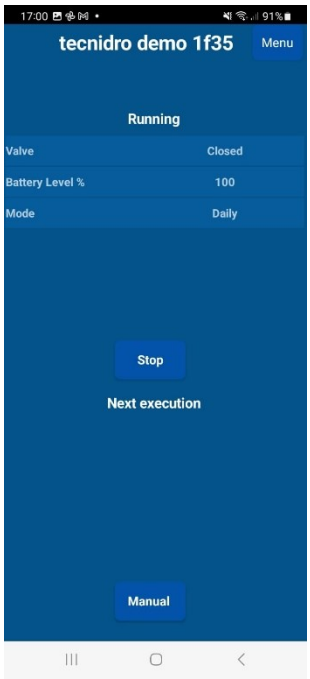
Where the status of the I-TIC device is shown

- Valve Status (Open/Closed)
- Battery Level % (100%, 67%, 33%, 0%)

The indication Device **STOPPED** indicates that the control unit is in stand-by, any programmes present are NOT running. It can be **ENABLED** with the Restart button

In this way, the control unit is activated and ready for the next settings





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The Stop button puts the control unit in stand-by mode

The Manual button allows manual opening

The Next Execution indication specifies the date and time of the next operation if scheduled operations are set.

With the control unit DISABLED all automatic operations related to PROGRAMMES or CYCLE mode are suspended.

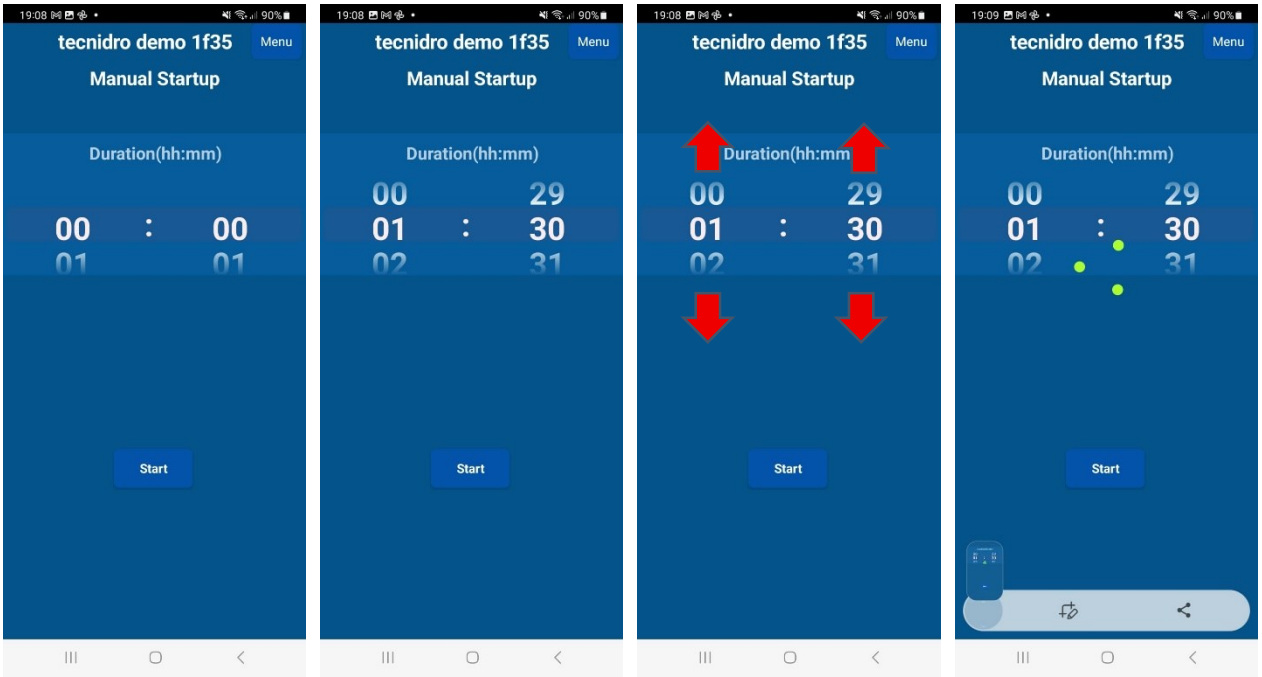
NOTE: When the I-TIC control unit is started or restarted (e.g. following battery replacement) it restarts in stand-by mode, i.e. with the device indication STOPPED, so it must be restarted as indicated in the previous section.

Furthermore, during the start/restart phase, a closing operation of the solenoid integrated or connected to the control unit itself is always carried out.

3.1 Manual opening

At any time (even with the control unit DISABLED) it is possible to perform a manual opening operation either with the Manual button at the bottom of the page or with the Manual button in the Menu list.

A window is then opened that allows you to select the duration of individual irrigation by setting the hours and minutes value via scrolling fields



Pressing the Start causes the valve to be opened and a page indicating the Open status and the remaining opening time is then displayed. Closing is possible at any time by pressing the Close button and confirming the closing.

During opening, 3 green dots may be visible rotating on the screen, these highlight the app's communication with the I-TIC device.

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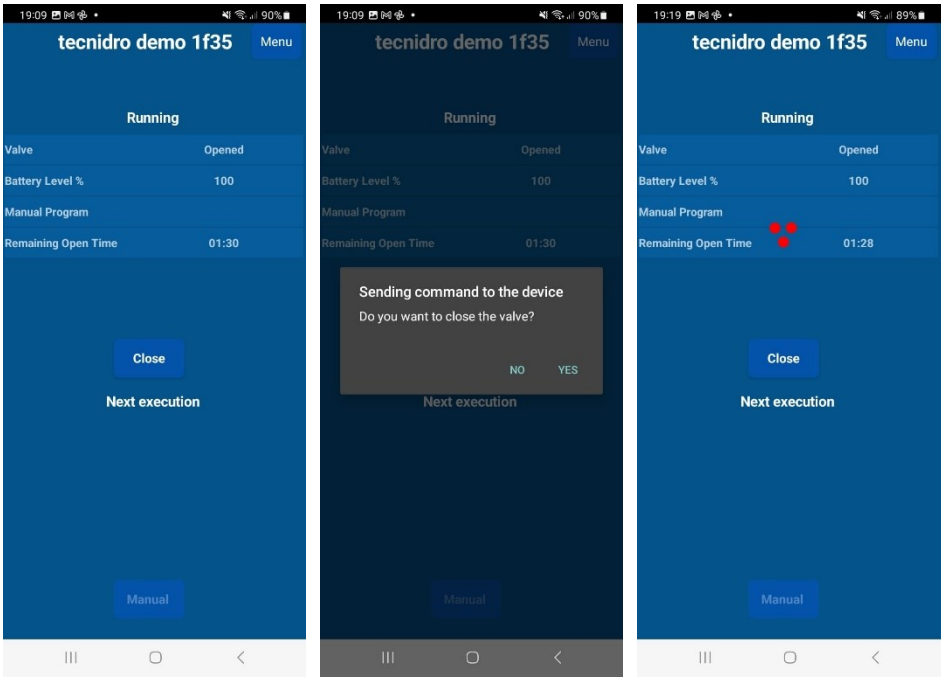
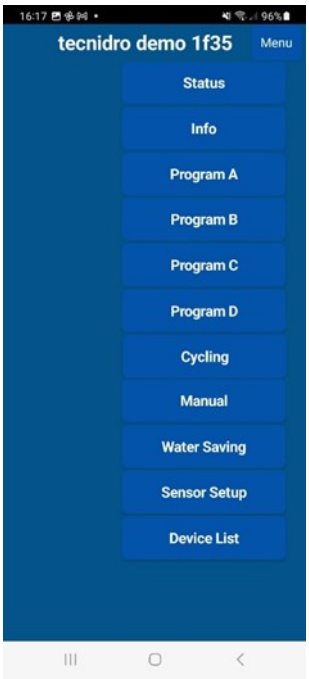


During closing, 3 red dots may be visible rotating on the screen, these highlight the communication of the APP with the I-TIC device.

3.2 Menu

Various activities can be performed from this page:

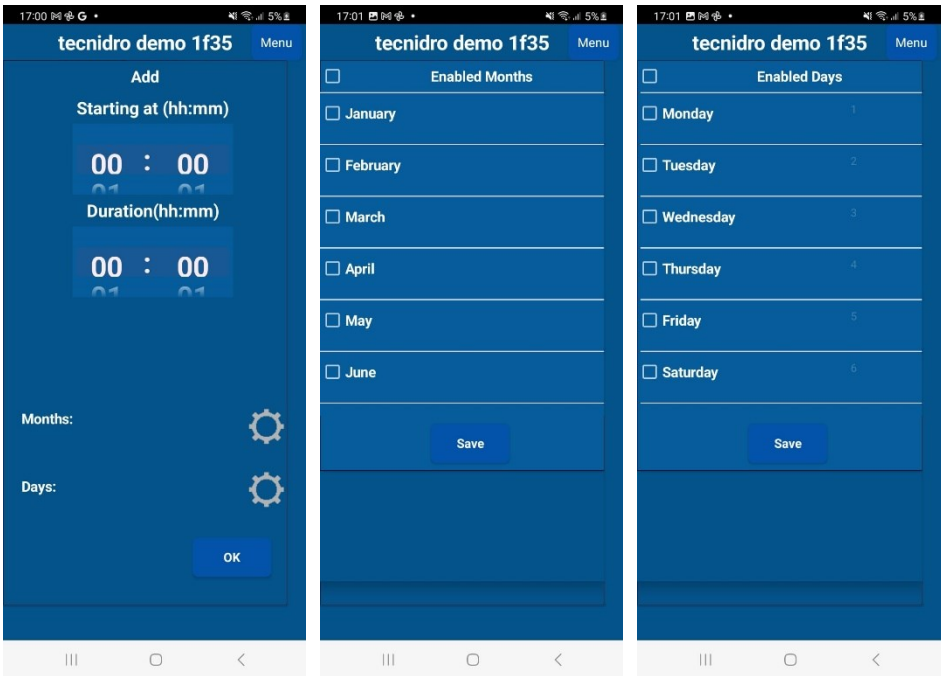


Status	The status of the device is displayed (valve status, battery level, device Active or in STOPPED mode).
Info	This page displays various device information (including PIN), as well as battery level, device FW version and APP version, from this page you can also delete all programmes (Clean Progs) and remove the device from the APP's device list (Remove).

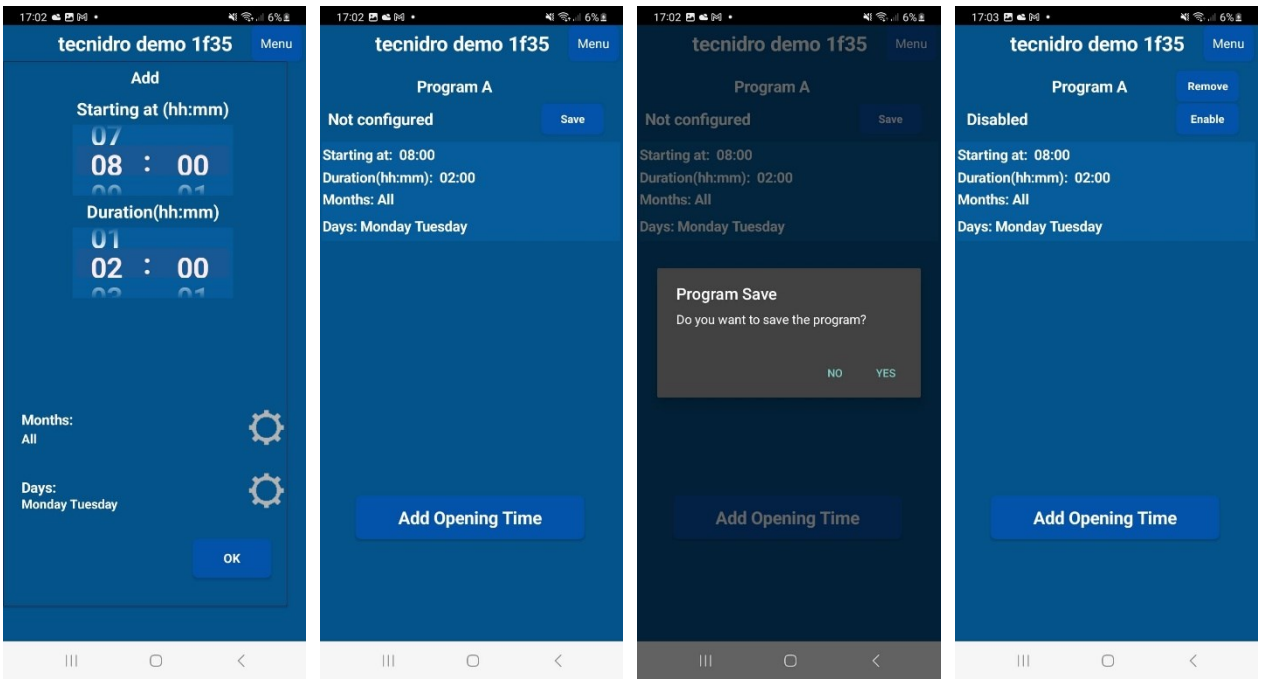
Program A, B C, D	These menus are used to set the weekly schedules, programs A,B,C,D are **alternative** programs to each other and are enabled individually (this is useful to pre-set different programmes to be adopted in different crop phases) It is possible to set (in total freely distributed among the four programs) up to 24 starts with a delivery duration of up to 24 hours each. It is also possible to specify the day(s) of activation as well as the month of activation for that specific departure.
Cycling	This menu allows cyclical programming of openings, 4 daily starts and a duration of up to 24 hours are available
Manual	Allows you to immediately perform a manual opening operation by setting its duration
Water Saving	This menu allows you to vary the duration of the sprays individually month by month from the set duration without changing individual programmes. It is possible to make a setting between 0 and 200%. Obviously, the default setting of 100% leaves the set durations unchanged.
Sensor Setup	This menu allows the setting of any counter connected to the I-TIC control unit and the enabling/disabling of the rain sensor.
Device List	Allows you to return to the list of I-TIC devices enabled on your smartphone/tablet

3.3 Weekly programming

In the following example, Program A was selected and a new program was created with an opening time of 08:00 for 2 hours for all months of the year and activated for Mondays and Tuesdays



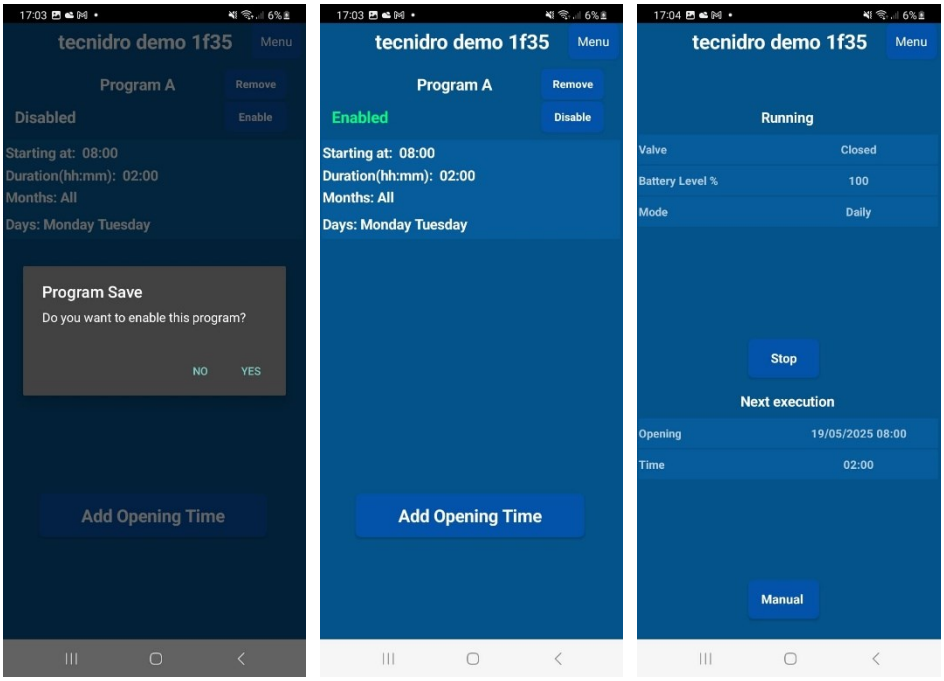
The first box next to Enabled Months and EnabledDays allows you to select all months/days



The Save button allows that programme to be saved on the device (all programmes are always saved on the device)

The Add Opening Time button allows you to add a new start to the programme, allowed a total of 24 starting's for the programs A, B, C and D)

Enable button enables that program and runs it on the device, the **Enabled** indication shows that that program is running. The program can be stopped with Disable button.



The Device Status page displays the date, time and duration of the next opening, and there is always a Stop button to interrupt all the running programme.

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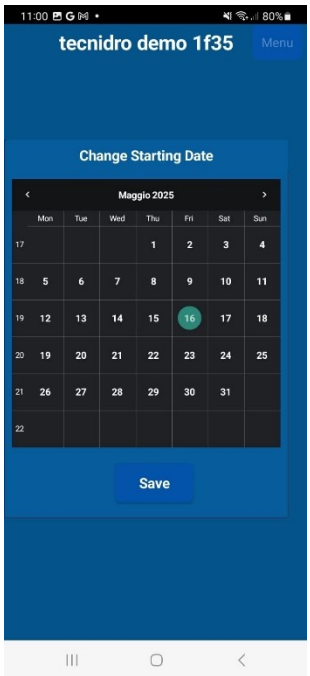
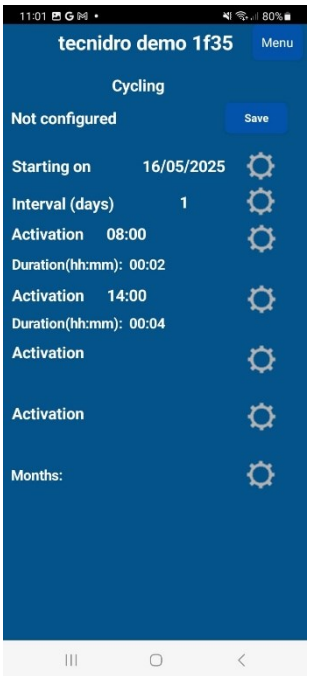
3.4 Cyclic Programming

For this setting, it is possible to define:

- Start date
- Interval in days between one opening cycle and the next one
- Time and duration of individual stats (up to 4)
- Activation months, you can select the months for which the valve is to be activated according to the cyclic schedule





The Save button saves the programme on the device, which can then be enabled with the Enable button

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3.5 Water Savings

On this page it is possible to set the “Water Saving” values individually for each month of the year, this automatically changes the duration of a programmed irrigation according to the set value between 0% and 200%, (in steps of 5%). Obviously, leaving the value 100% set, the durations set in the programs remain unchanged.

This feature can be useful for dealing with any critical or, conversely, particularly rainy periods by raising or lowering the irrigation duration overall without having to intervene individually on each setting.

Selecting 100% sets the value of all months to 100%.

3.6 Sensor Setting

From this page it is possible to configure the presence of meter, equipped with a pulse output, connected to the I-TIC device. It is possible to set the volume of water corresponding to each pulse received from the counter to the values.

- 1 pulse = 10 litres
- 1 pulse = 100 litres
- 1 pulse = 1000 litres (1 cubic meter)
- 1 pulse = 10000 litres (10 cubic meters)

On the same page, the rain sensor can be activated/deactivated.

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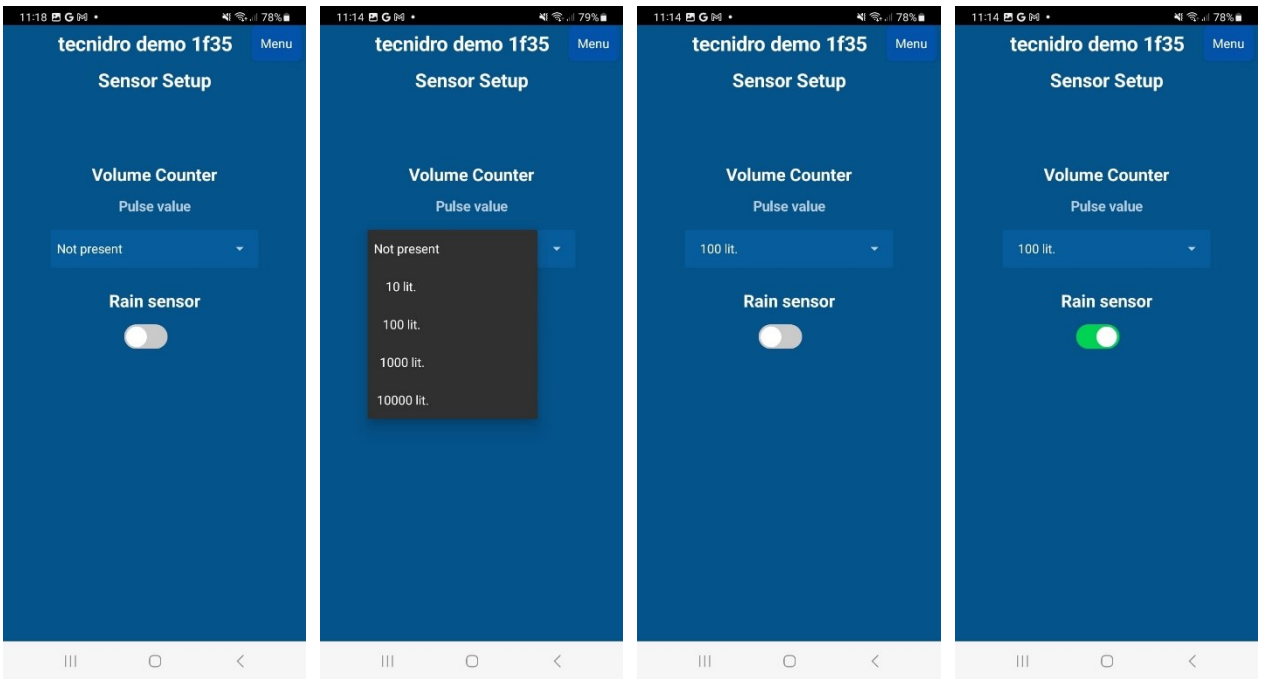
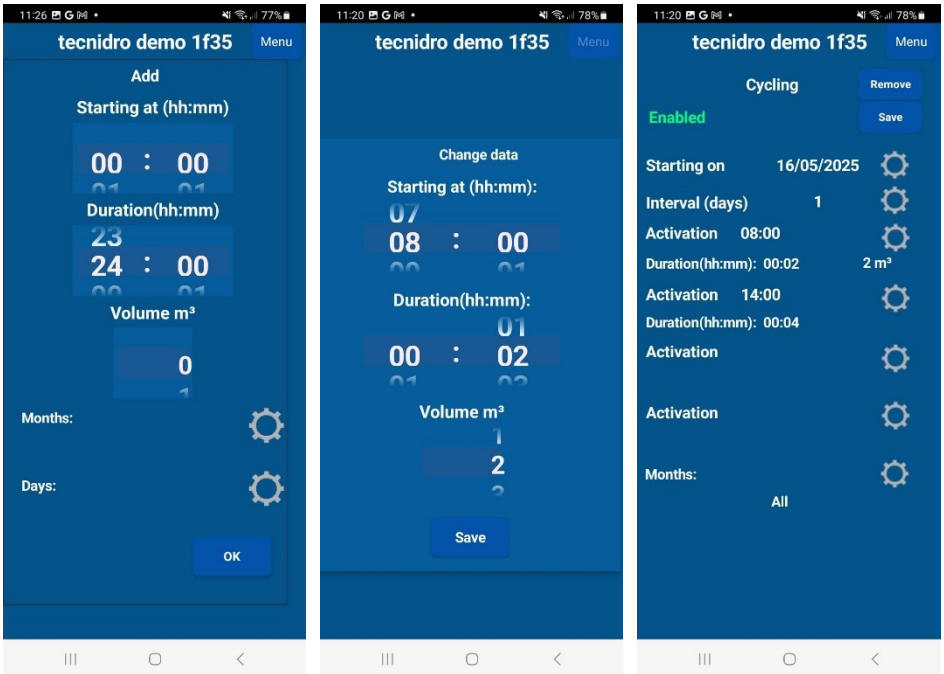


NOTE: By enabling the volume counter in addition to the watering duration it is also possible to set the volume of water to be dispensed for each manual or automatic watering. Watering will then end when the set time or volume is reached (the first value that is reached).

With the counter activated in the openings setting, the duration value is displayed on the value 24:00 hh:mm, of course it will have to be set to the desired value.

In the volume setting it is possible to select the number of cubic metres with an integer value between 0 and 99 mc (with the setting 1 pulse =10 mc it is possible to set between 0 and 90 mc in 10 mc intervals).

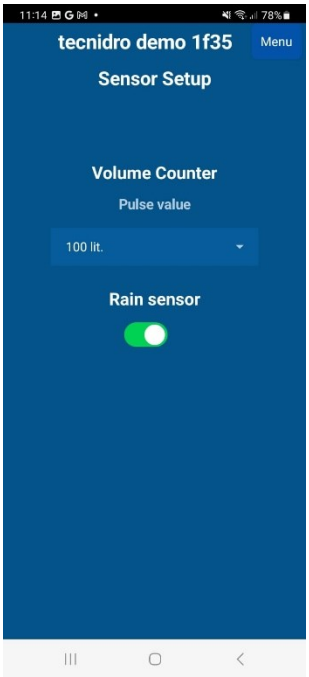


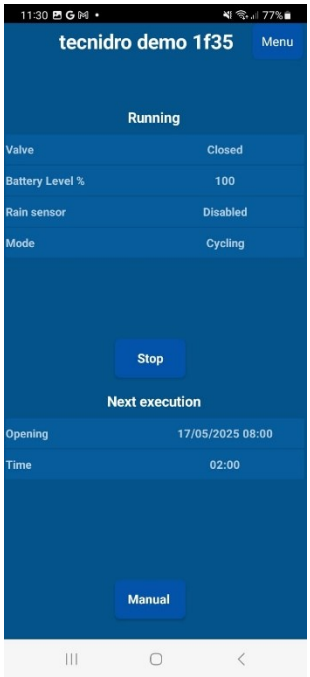
NOTE: The presence of the counter input is an optional feature for the I-TIC device, if it is equipped with one, there are two outgoing wires (white and grey normally) for meter connection.

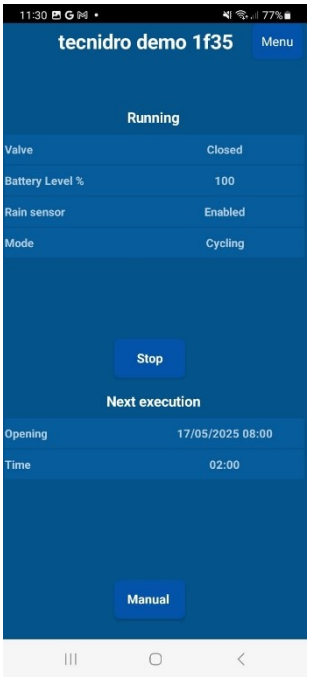
3.7 Rain sensor

The abilitation of the rain sensor can also be seen on the status page, where there is a new status line referring to the rain sensor.

- **Disabled** indicates that the sensor is activated (from the sensor setup page) and any rain situation is detected
- **Enabled** indicates that the sensor is activated, and a rain situation is detected at that moment







NOTE: From a connection point of view, the rain sensor corresponds to the continuous yellow wire (input/output) on the device, the rain sensor can be connected to the I-TIC device by cutting the yellow wire and connecting its ends to the sensor itself, which typically has these conditions.

- contact closed rain sensor disabled
- open contact rain sensor enabled

Should the rain sensor be disconnected from the device, it is recommended to restore the continuity of the yellow wire by joining its two ends.

The rain sensor only intervenes on automatic activations set with the weekly or cyclic programmes as follows.

1. When activated, the control unit constantly monitors the status of the rain sensor
2. When the valve is supposed to open, if the rain sensor is ENABLED the opening is not carried out
3. If, however, during the expected opening time the sensor returns DISABLED, then the valve is opened for the remaining time.
4. During irrigation, i.e. when the valve is open, if the rain sensor becomes ENABLED then irrigation is stopped
5. If during the remaining watering time the rain sensor becomes DISABLED again, then watering is resumed.

NOTE: Be careful NOT to activate the rain sensor on I-TIC devices without this feature, this may result in failure to perform the programmed automatic operations.

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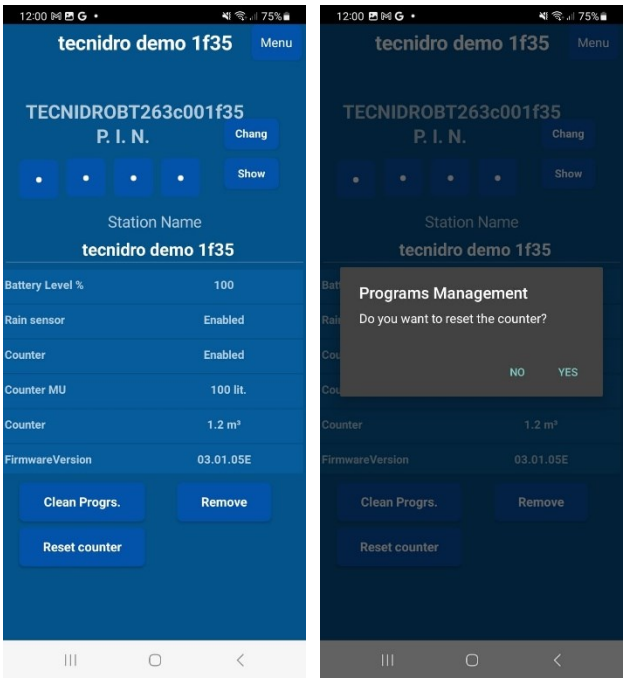
3.8 Water meter

When the counter is enabled, additional information about the counter, the settings value and the counter totalised value are displayed on the Info page of the Menu.

As previously indicated for each opening operation the volume to be delivered can be set.



This totaliser can be reset using the Reset counter button

NOTE: from a connection point of view, the I-TIC is optionally available with meter wiring, this must be specified when ordering/purchasing.

4 Notification of anomalies

The app notifies the device page of any anomalies presented during operation such as

- Valve error
- Low battery

NOTE: the “Valve error” the indication specifies that the opening or closing operation was not performed, typically this may be due to a battery problem.

The valve error indication remains present until a subsequent successful operation on the valve.

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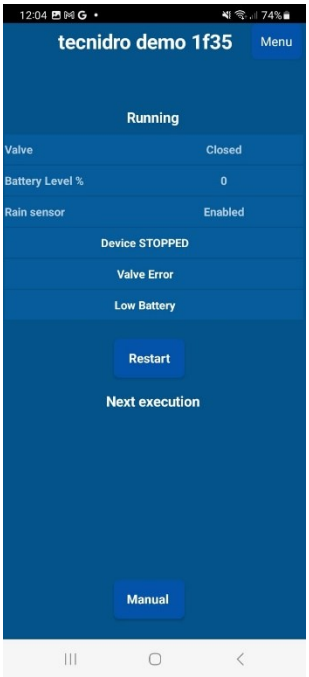
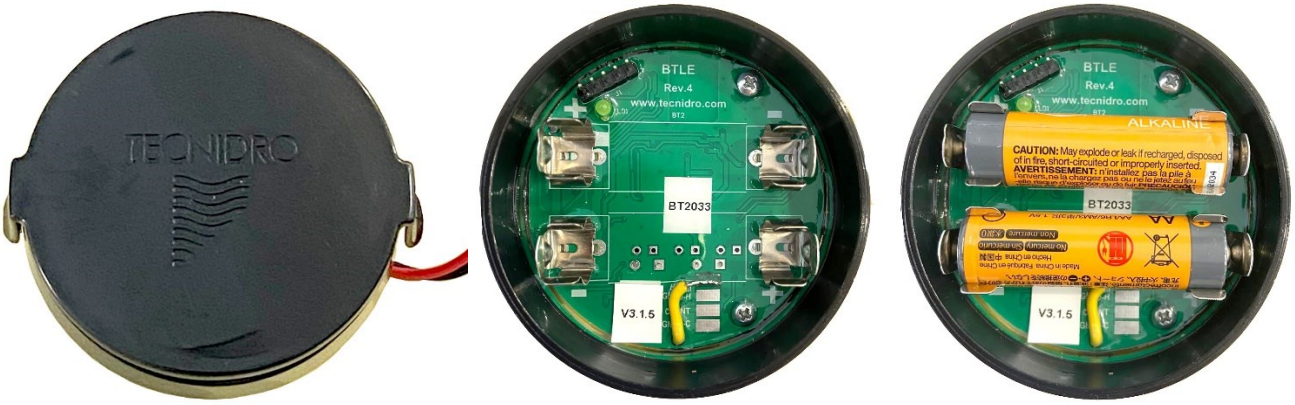


5 Battery

The I-TIC device foresees No.2 AA size batteries (LR6) 1.5V .

These are inserted and locked by special clips, take care at polarity as indicated.



It is recommended to use good quality alkaline batteries that will allow the device to operate for a full irrigation season / 1 year.

The device notifies the battery status with the following indications - Battery%

- 100% new battery fully charged
- 67% battery still in good condition
- 33% battery nearing discharge, prepare to replace
- 0% battery low, to be replaced, device is disabled, automatic programs are STOPPED

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5.1 LED Indications

At battery replacement and device startup the following indications are visible on the led inside the device.

Approximately n.10 flashes at a rate of about one second for device startup and n.2 rapid flashes for Bluetooth initialization

A close command is also given to the connected valve to put the system in a CLOSED condition

When the device is started, it is in the STOPPED state.

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